

Title: Bio

Grade Level: 10

Subject: Biology

Difficulty: MEDIUM

Student Name: _____

Marks: 56 Pts

Time: 85 Mins

Roll No: _____ **Section:** _____

SECTION A: MULTIPLE CHOICE QUESTIONS

Instructions:

Choose the best matching option for each question.

1. What is the process by which plants convert sunlight into energy? [1 Marks]

2. Which organelle is responsible for cellular respiration? [1 Marks]

3. What is the term for the movement of molecules from an area of high concentration to an area of low concentration? [1 Marks]

4. Which type of tissue is composed of muscle cells? [1 Marks]

5. What is the primary function of the mitochondria in a cell? [1 Marks]

SECTION B: SHORT QUESTIONS AND DIAGRAM-BASED QUESTIONS

Instructions:

Answer each question in detail and draw diagrams as required.

1. Describe the structure of a plant cell. [2 Marks]

2. Draw and label a diagram of the human circulatory system. [5 Marks]

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3. Explain the process of photosynthesis, including the reactants and products involved. [2 Marks]

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4. Draw and label a diagram of the structure of an animal cell. [5 Marks]

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5. Describe the role of the nervous system in the human body. [5 Marks]

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6. Draw and label a diagram of the process of mitosis. [5 Marks]

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7. What are the functions of the kidneys in the human body? [2 Marks]

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SECTION C: NUMERICAL PROBLEMS

Instructions:

Solve each numerical problem and provide your answer in the correct format.

1. A population of bacteria grows from 100 to 1000 in 5 days. What is the rate of growth per day? [5 Marks]

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2. A car travels from city A to city B at an average speed of 60 km/h. If the distance between the cities is 240 km, how long does the journey take? [5 Marks]

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3. A certain species of bacteria doubles in number every hour. If there are initially 200 bacteria, how many will there be after 5 hours? [5 Marks]

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4. A group of friends want to share some money equally. If they have \$150 to share and there are 15 friends, how much money will each friend receive?

[5 Marks]

5. A ball is thrown upwards from the ground with an initial velocity of 20 m/s. If the acceleration due to gravity is 9.8 m/s^2 , how long does it take for the ball to reach its maximum height?

[5 Marks]
